IMPLEMENTATION APPROACH

PROBLEM UNDERSTANDING

Organizations receive large volumes of documents every day, including identity proofs (Aadhaar, PAN, Passport, etc.), financial statements, invoices, receipts, legal agreements, tax documents, compliance records, and business registration documents. Manually identifying, sorting, and managing these documents is slow, error-prone, and heavily dependent on human effort. This leads to delays in processing, compliance checks, and overall data management.

Existing workflows often rely on manual tagging, inconsistent folder structures, or basic upload forms with no smart classification or search capabilities. As a result, it becomes difficult to quickly find specific documents (for example, “all Passport uploads” or “invoices from a particular vendor”), and there is no systematic way to detect or flag potentially restricted or sensitive content.

ParseFlow Doc Classifier addresses this by building an intelligent document-classification system that automatically reads uploaded documents and classifies them into the correct category. The system supports multiple formats (PDF, images), provides fast AI-driven classification, and stores documents in a structured, searchable vault. This reduces manual effort, speeds up processing and compliance workflows, and enables seamless API integration with enterprise systems for scalable document management.

SOLUTION WORKFLOW

ParseFlow Doc Classifier follows the exact workflow shown in your diagram, with a clear 3-tier classification cascade and category-based storage:

User Login & Document Upload

User logs in and uploads a PDF or image document via the web interface.

Supported formats include identity proofs, financial docs, legal agreements, compliance records, tax documents, and business registration docs.

Tier 1: ML Classifier – High Confidence Check

ML Model immediately analyzes the document for identity proofs and other high-confidence patterns.

If confidence ≥ threshold (e.g., Passport at 99.6%):

→ Instant Final Result → Skip to storage.

If confidence < threshold: Proceed to Tier 2.

Tier 2: Vision-LLM Fallback

Vision-LLM processes complex or ambiguous documents (e.g., Aadhaar, PAN cards).

Extracts key fields (name, ID number, DOB) and refines document type classification.

If successful: Generate final classification.

If Vision-LLM fails: Proceed to Tier 3.

Tier 3: EasyOCR Fallback

EasyOCR acts as the ultimate fallback for edge-case layouts, poor scans, or unusual format

Extracts readable text and basic patterns to enable classification.

Guarantees no document is left unprocessed.

Rich JSON Output & Category Storage

System generates structured JSON output:

json

```
{
  "document_type": "Passport",
  "confidence": 0.996,
  "key_fields": {"name": "...", "id_number": "..."},
  "folder": "Identity/Passport"
}
```

HIGH LEVEL SYSTEM ARCHITECTECTURE

1. Frontend Layer (React)

Clean upload interface with drag-and-drop support

Real-time confidence gauges and processing status

Push notification system for live upload awareness

Dual chatbot interface (GuideBot for navigation, DocBot for search)

Folder view showing Identity/Financial/Legal/Compliance/Tax/Business categories

2. Backend Layer (Django REST API)

upload endpoint receives documents and triggers classification

3-Tier processing pipeline

ML Classifier → [High Confidence?] → Instant Result

↓ (Low Confidence)

Vision-LLM → [Success?] → Structured Output

↓ (Failure)

EasyOCR → Final Fallback → Basic Classification

File storage for original PDFs/images

JSON response with {type, confidence, key_fields, folder}

3. AI Classification Engine

Tier 1: ML classifier optimized for identity proofs (Passport 99.6%)

Tier 2: Vision-LLM for complex layouts and field extraction (name, ID, DOB)

Tier 3: EasyOCR fallback for edge cases and poor scans

4. Data Layer (MongoDB)

text

```
{
  "doc_id": "uuid",
  "original_file": "path/to/upload.pdf",
  "document_type": "Passport",
  "confidence": 0.996,
  "key_fields": {"name": "John Doe", "id": "Z1234567"},
  "category_folder": "Identity/Passport",
  "timestamp": "2026-03-27T...",
  "user_id": "team_xyz"
}
```

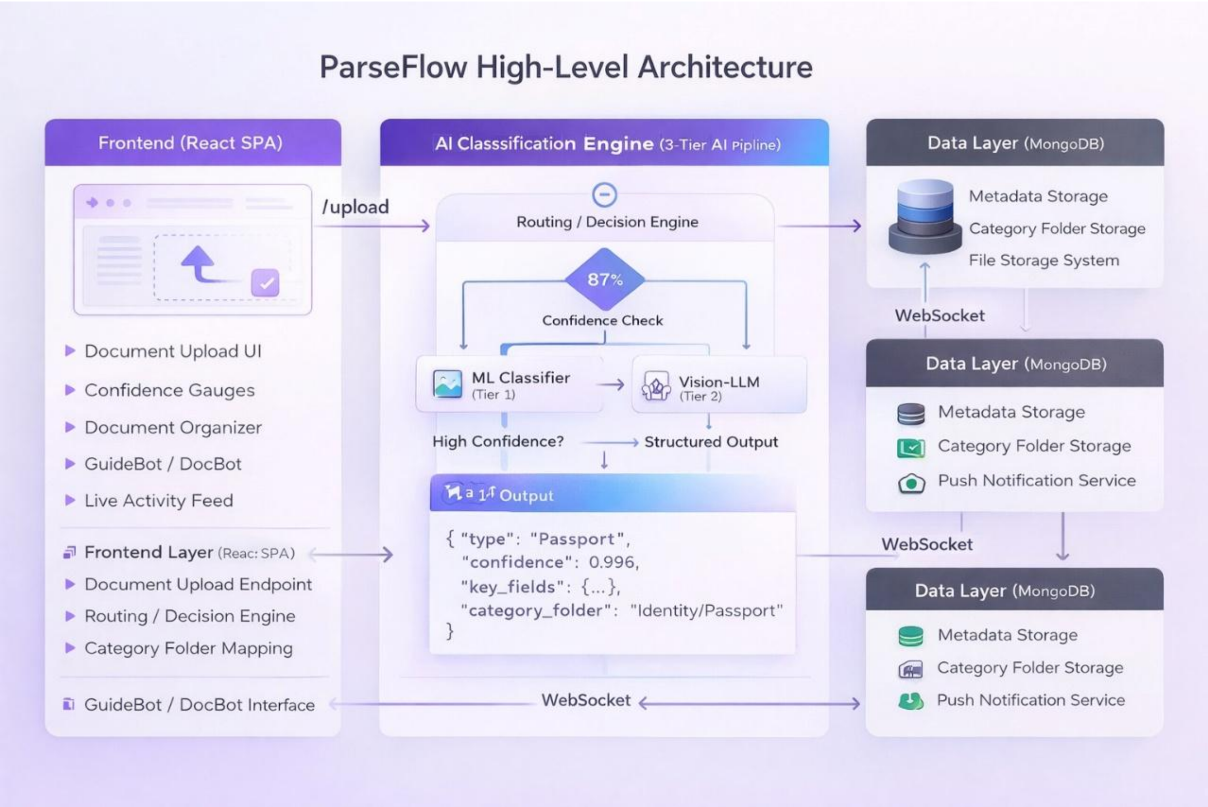
5. Integration Layer

Clean REST API for external systems

CSV/JSON export endpoints for audit trails

WebSocket for real-time push notifications

WORKFLOW



CHALLENGES

1. Balancing Speed vs. Accuracy

Challenge: Vision-LLM is accurate but slower than ML classifier; EasyOCR adds latency for edge cases.

Solution: 3-tier cascade prioritizes ML for high-confidence cases (99.6% Passport → instant), only escalates complex docs.

2. Diverse Document Layouts

Challenge: Aadhaar/PAN cards, handwritten receipts, and cropped images have inconsistent layouts.

Solution: Vision-LLM handles complex layouts + EasyOCR fallback ensures 100% coverage.

3. Low-Quality Scans & Photos

Challenge: Blurry phone photos or poor scans fail basic OCR.

Solution: Pre-processing pipeline + EasyOCR's robust text recovery for edge cases.

4. Sensitive Document Privacy

Challenge: Identity proofs contain PII (name, Aadhaar number, DOB) that must be protected.
Solution: Store only structured metadata in MongoDB; encrypt raw files separately with user/team access controls.

5. Per user memory limitations

Challenge: MongoDB storage for each user's document metadata, thumbnails, and search indexes grows quickly with frequent uploads.

Solution: Implement storage quotas per user/team + automatic archiving of older documents to external storage.

6. Category Edge Cases

Challenge: Documents that don't fit standard categories (Identity/Financial/Legal/etc.).
Solution: "Others" catch-all folder + manual override option for organizers.